Tiyasa Saha

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Summary:

Data professional with experience building analytics solutions and machine learning models to solve complex business problems. Skilled in Python, SQL, and data pipelines, with a strong focus on translating data into actionable insights for decision-making. Experienced in working with large datasets, stakeholder collaboration, and applying statistical and machine learning techniques to improve efficiency, reduce risk, and support strategic initiatives.

Skills:

Data Analysis & Analytics: KPI Tracking, Statistical Analysis, Data Validation, Business Insights

Database Management:
- **RDBMS**: PostgreSQL, MySQL, SQLite, SQL Server 2014
- **NoSQL**: MongoDB, Cassandra
- **Tools**: PL/SQL Developer, pgAdmin, SQL Loader

ETL and Data Warehousing (DWH):
- **Tools**: Apache Airflow
- Expertise in ETL pipelines and managing data in DWH environments

Data Visualization:
- **Tools**: Tableau, D3.js, Power BI
- Dashboards and storytelling using Tableau Desktop and D3

Web Technologies: HTML, CSS, JavaScript

Programming Languages: Python, SQL, Java

Machine Learning & Statistical Techniques: Regression, Clustering, SVM, Decision Trees, Classification, Recommendation Systems, Association Rules, Predictive Analysis

Cloud Platforms:
- **Amazon Web Services (AWS):** EC2, S3, Redshift, Azure Machine Learning

AI & NLP: LLMs, RAG, LangChain, OpenAI APIs, Prompt Engineering, OpenAI SDK

Deep Learning Frameworks: TensorFlow, Keras, Torch

Other Tools and Technologies: Git

Experience:

Data Engineer | Saayam for All, San Jose, CA — **Mar 2026 - Present**
- Built and deployed scalable machine learning models using Azure Machine Learning (AML) to optimize intelligent matching of beneficiaries with volunteers and resources.
- Architected distributed data processing pipelines leveraging Apache Spark and diverse databases (SQL, NoSQL) to handle large-scale, complex datasets, including user profiles, donation records, and global help requests.
- Applied Natural Language Processing (NLP) and machine learning algorithms (Random Forests, K-means, Naive Bayes) in Python to analyze and parse incoming text requests, accurately categorizing and routing human needs.
- Developed and deployed predictive models via Python APIs to segment donor behavior and volunteer engagement, maximizing platform outreach, retention, and support efficiency.
- Designed interactive visualizations and dashboards using Tableau and Power BI, while conducting rigorous A/B testing and statistical validation (PCA, K-fold) to drive strategic resource allocation and internal decision-making.
- Collaborated with stakeholders to translate business needs into analytical solutions and improve decision-making processes.

AI Intern | Interview Query (Remote) — **Sept 2025 - Mar 2026**
- Built data pipelines to process high-volume user telemetry and support real-time analytics workflows.
- Applied NLP and machine learning (SVM, Decision Trees) via scikit-learn to automate the parsing and categorization of interview question text and candidate code submissions.
- Developed predictive models (Random Forest, Logistic Regression) to segment user behavior, accurately forecasting premium subscription conversions and personalizing practice recommendations.
- Optimized A/B testing and built interactive Tableau/Power BI dashboards to visualize user engagement and retention metrics, driving data-driven platform decisions.
- Leveraged PySpark (MLlib) and Azure Databricks for large-scale feature engineering, improving candidate success predictions through rigorous model evaluation (ROC, AUC).

System Engineer | Tata Consultancy Services (TCS), Kolkata, India — **Oct 2020 - Aug 2023**
- Built and maintained scalable data pipelines using Python and SQL to process and transform large datasets, improving data availability and reducing system latency by 30%.
- Supported analytics and modeling workflows by preparing clean, structured datasets and performing feature engineering for downstream business and analytical use cases.
- Managed the end-to-end machine learning lifecycle from initial client requirements gathering and conceptual data modeling to feature engineering and production deployment using Python and statistical techniques.
- Designed and optimized Tableau dashboards by fine-tuning data connections, and automating background tasks, delivering actionable insights to business stakeholders.

Education:

MS in Data Science: University of Massachusetts Dartmouth, Dartmouth, MA, USA	**4.0/4.0 GPA**	**Sept 2023 - Aug 2025**
B-TECH in Computer Science: Amity University Kolkata, West Bengal, India	**7.98/10 CGPA**	**Aug 2016 - Sept 2020**

Projects:

Breast Cancer Chatbot (BrCaBot) (GitHub Code) (Site Link): — **Jan - May 2025**
- Tools: Python | Flask | OpenAI | LangChain
- Built and deployed a Generative AI application using a Retrieval-Augmented Generation (RAG) pipeline to answer real-time patient queries.
- Designed prompt strategies and retrieval workflows to improve response accuracy and reliability.
- Integrated OpenAI APIs with a Flask backend to enable scalable, low-latency inference.
- Explored trade-offs between response quality, latency, and retrieval performance during experimentation.

BRCA (Breast Invasive Carcinoma) Treatment Predictor (GitHub Code): — **Sept - Dec 2024**
- Tools: XGBoost | t-SNE | Logistic Regression
- Developed predictive models to analyze clinical data and identify patterns in treatment outcomes, achieving over 90% accuracy.
- Performed feature engineering, hyperparameter tuning, and cross-validation to ensure model generalization.

Statistical Analysis of Public Data (Site Link): — **Jan - Apr 2024**
- Tools: Decision Trees | Naïve Bayes | K-Means | KNN
- Developed supervised and unsupervised models on national incident data; achieved >85% accuracy in classification, resulting in the identification of previously undiscovered correlations in incident patterns.

US Incident Time Series Dashboard (GitHub Code) (Site Link): — **Sept - Dec 2023**
- Tools: D3.js | HTML/CSS | ARIMA
- Developed an interactive time series dashboard visualizing 8 years of U.S. incident data using ARIMA forecasting to analyze long term incident trends and seasonal effects.